FMS 304 – RESEARCH METHODOLOGY

PSALMDATA POP Examination Revision Summary | Official NOUN study material (2023 edition) | Past-question sessions: 2022_2, 2023_1, 2024_1, 2024_2, 2025_1 and 2025_2.

This revision note follows the exact three-module, fifteen-unit order of the supplied FMS 304 material. Past questions appear beside the topics they test. Repeated questions are linked to their sessions; the answers are condensed for quick POP revision rather than presented as lengthy textbook chapters.

MODULE ONE

UNIT 1: CONCEPTUAL FRAME WORK OF RESEARCH

1.0 Introduction

Research begins when a question needs a dependable answer. Unlike casual information searching, it follows a planned process of studying evidence and reaching conclusions about a clearly stated problem.

3.1 The Concept of Research

Research is the systematic collection, analysis and interpretation of data to increase understanding or solve a problem. Leedy describes it as a data-supported search for an answer; Travers links research to an organised body of scientific knowledge used to explain, predict and sometimes control events. It is **not** mere copying, moving facts from one source to another or indiscriminate information gathering.

Eight process features identified by Leedy: a research question or problem; a clear goal; a definite procedure; division of a large problem into smaller ones; guiding questions or hypotheses; stated critical assumptions; collection and interpretation of data; and a cyclical or helical process in which findings generate new questions.

PAST QUESTION - 2025_2 Q1(a) – 10 marks

Define the concept of research and discuss its significance in everyday life.

Answer: Research is a systematic, objective, evidence-based inquiry that gathers and analyses data to answer questions, solve problems and improve understanding. In everyday life, it helps people identify real problems, make decisions on reliable information, test claims rather than accept guesswork, improve products or services, predict likely outcomes and generate new knowledge. For example, a hotel can investigate why customers complain before changing its service system.

3.2 Characteristics of Research

Objectivity: suspend personal preferences and let evidence guide conclusions. **Precision:** employ clear concepts, dependable measurements and exact language. **Design:** state the problem and hypotheses, select procedures, gather and analyse data, test conclusions and report the findings. **Verifiability:** allow others to scrutinise, reanalyse or replicate the investigation. Other qualities identified in the text include empirical evidence, careful recording, logical reasoning, expertise, patience, openness to validation and sometimes courage.

PAST QUESTION - 2025_2 Q1(b) – 10 marks

Explain the characteristics of research, providing examples to support your points.

Answer: The four major characteristics are (1) objectivity: a researcher reports the actual responses instead of selecting those that favour a personal belief; (2) precision: terms such as random sampling and reliability are defined and results reported clearly; (3) design: a study has orderly steps from problem identification to reporting; and (4) verifiability: another scholar can reanalyse the data or repeat the study on a different sample. Sound research is also empirical, systematic, logical, carefully documented and open to criticism.

3.3 Types of Research

Fundamental/basic research develops or evaluates theories and broad knowledge without primarily seeking an immediate practical remedy. **Applied research** tackles a particular problem in a real setting. The distinction is not absolute: theory can solve practical problems and applied findings can strengthen theory. **Quantitative research** measures variables numerically and uses statistics to test relationships or predictions. **Qualitative research** studies meaning, participants' experiences and events in natural settings, presenting rich descriptions in words.

Other categories named in the course include **action, case/field study, correlational, descriptive survey, ethnography, ex post facto/evaluation, grounded theory, historical, phenomenological, quasi-experimental** and **true experimental research**. Their differences lie mainly in purpose, setting, data and degree of control.

PAST QUESTION - 2025_2 Q1(c) – 5 marks

Differentiate between fundamental and applied research, giving examples of each.

Answer: Fundamental research aims at extending knowledge and developing theory; for instance, investigating general principles of customer motivation. Applied research aims at solving an identified, immediate problem; for instance, finding why a particular hotel's guests are dissatisfied. Basic research is knowledge-oriented, while applied research is problem-oriented, although their findings may complement one another.

PAST QUESTION - 2024_2 Q1(a) – 8.75 marks

Briefly write short notes on (i) case and field study research (ii) action research (iii) correlational research.

Answer: (i) Case and field study research examines an individual, group or institution closely in its natural environment, using several sources of information. (ii) Action research is applied, local problem-solving inquiry in which an action is planned, tried and evaluated. (iii) Correlational research uses statistical techniques to investigate the degree of relationship among variables; association alone does not establish cause and effect.

3.4 Purpose of Research

The material presents **three purposes in a hierarchy: (1) training in research, (2) problem-solving and (3) search for truth or knowledge generation**. Training builds the skills to design a study and analyse evidence. Problem-solving diagnoses practical difficulties and tests remedies. The search for truth develops broader generalisations, principles and theories from repeated observations and scientific tests.

PAST QUESTION - 2025_2 Q2 – 15 marks

Discuss the THREE main purpose of research in a hierarchical or taxonomic fashion.

Answer: The purposes are (1) training in research: developing the knowledge, scientific attitude and skill needed to identify problems, select designs, gather data and draw defensible conclusions; (2) problem-solving: investigating practical difficulties in an organisation or community and finding evidence-based improvements; and (3) search for truth or knowledge generation: using systematic observations, testing and generalisation to develop principles and theories that explain events beyond one immediate case. The hierarchy moves from building the researcher's competence, through applying inquiry to specific problems, to extending reliable knowledge.

PAST QUESTION - 2024_2 Q1(b) – 8.75 marks

Scientific approach has series of six (6) steps, which may vary from one author to another, mention the steps.

Answer: The material lists six steps: (1) observe a phenomenon; (2) recognise and clarify the problem situation; (3) derive a more or less formal hypothesis; (4) develop a design for testing the hypothesis; (5) verify or refute the hypothesis; and (6) subject the results to further tests and refinement. Review, measurement, manipulation of relevant factors and data analysis support these steps.

Quick Revision: **Research = problem + planned inquiry + data + reasoned conclusion.** Remember the four major characteristics, the basic/applied distinction and the three purposes.

UNIT 2: SKILLS OF SCIENTIFIC INVESTIGATION

1.0 Introduction

Scientific investigation uses observation, logical explanation, hypotheses and evidence to make a problem clearer. It differs from relying only on authority, custom or personal experience.

3.1 The Scientific Method and Activities

Science aims to describe phenomena accurately, explain conditions under which they occur and organise evidence so relationships can be understood and events predicted. The course identifies four assumptions: **order** (nature follows discernible patterns), **determinism** (events have identifiable antecedents), **parsimony** (prefer adequate simple explanations) and **empiricism** (test ideas against observable evidence). **Replication** checks whether findings recur under comparable conditions.

PAST QUESTION - 2024_1 Q1(a) – 8.75 marks

Scientific inquiries involves four (4) assumptions which are represented by certain key words, briefly mention and explain these assumptions of scientific inquiries.

Answer: Order assumes that events in nature occur in discoverable patterns. Determinism holds that events have causes or preceding conditions that can be investigated. Parsimony favours the simplest adequate explanation instead of needless complexity. Empiricism requires systematic observation, measurement and experimental evidence to assess claims. These assumptions make scientific inquiry systematic, testable and capable of repetition.

3.1.1 Scientific Activities

The source describes: identify and define a problem; review earlier experience, findings and data; formulate tentative hypotheses; examine their logic and testability; collect additional evidence by observation, measurement or experiment; and classify, analyse and summarise findings into generalisations or principles. Other sources of knowledge named are **custom/tradition, authority, personal experience** and **syllogistic reasoning**, but scientific inquiry places special weight on verifiable evidence.

3.2 Facts and Theories

Facts are observations or findings obtained from empirical inquiry. A **theory** is a related set of concepts, definitions and propositions that explains and predicts phenomena by specifying relationships among variables. Facts provide the basis for theories; theories help organise facts, guide fresh observations and acquire meaning only when tested against evidence.

3.3 Hypothesis and Theory

A **hypothesis** is a tentative proposition explaining a limited set of facts or predicting a relationship and is available for testing. A **theory** is a wider explanatory framework supported by organised evidence; scientific **laws** have wider scope in the course's presentation. No theory should be regarded as beyond correction: new facts may require its modification or rejection.

PAST QUESTION - 2024_1 Q1(b) – 8.75 marks
Provide a clear distinction between hypothesis and theory.

> **Answer: A hypothesis is a provisional, specific and testable explanation or prediction about variables; it directs a particular investigation. A theory is a broader, organised system of concepts and propositions explaining relationships across many facts. A researcher tests a hypothesis with evidence; a theory integrates findings and generates further hypotheses. Both remain open to revision when new evidence appears.**

3.4 Purpose of Theory

A theory **organises existing knowledge, provisionally explains observed events and relationships, permits prediction** and stimulates further investigations. It helps a researcher identify which variables and evidence deserve attention.

3.5 Developing a Theory

The researcher collects facts, examines possible relationships, advances hypotheses, draws deductions and checks them against further facts. Wider generalisations gradually form a theoretical framework. A **construct**, such as motivation, may not be directly visible although its effects can be investigated.

3.6 Levels of Theorizing

The six levels in the course are **(1) hypothesis formation, (2) elementism, (3) descriptive theories and taxonomies, (4) classification, (5) postulates/incomplete theories and (6) fully developed axiomatic theories**. The first levels deal with relatively limited explanation; the highest use interconnected basic statements or axioms. The book groups levels five and six together as its 'higher levels'.

Quick Revision: **Order, determinism, parsimony, empiricism**; facts support theory, while theory organises facts.

UNIT 3: METHODS OF DATA COLLECTION IN RESEARCH - 1

1.0 Introduction

Collect only the information needed to answer the research questions or test the hypotheses. The type of data and the instrument used must fit the research design.

3.1 Data Collection in Research

Data are verbal or numerical information from which conclusions may be drawn. **Quantitative data** occur as counts or measurements; **qualitative data** describe qualities, meanings and experiences. Data collection must be relevant, systematic and tied to the study's variables.

3.2 Types of Data Used in Research Investigations

Primary data are first-hand information originated by the researcher for the investigation, such as interviews, questionnaires and direct observations. **Secondary data** are materials already gathered or recorded by others, such as books, journals, government records, earlier reports and published statistics. The key difference is **who originally obtained the information and for what purpose**.

PAST QUESTION - 2025_1 Q1(a) – 12.5 marks; 2023_1 Q3(b); 2022_2 Q1(D) – 5 marks
Identify and discuss the two main types of data used in research; distinguish between primary and secondary data.

Answer: Primary data are original, first-hand facts collected specifically for the present study through interviews, questionnaires, direct observations or measurements. They fit the immediate research question but require time and resources to obtain. Secondary data are previously existing facts compiled by another person or institution, including journals, books, census reports, records and earlier studies. They save time and may provide broad coverage, but their relevance and reliability must be evaluated. The distinction concerns the origin of the information, not whether it is numerical or verbal.

3.3 Questionnaire Method of Data Collection

A **questionnaire** is a carefully prepared form of questions answered by respondents, normally without the researcher's explanation during completion. It is useful in survey research with sizeable samples. Each item must generate information relevant to the research objectives and hypotheses.

PAST QUESTION - 2023_1 Q4
Most researchers who conduct survey researches make use of the questionnaire as the instrument for data collection. Substantiate this statement.

Answer: Questionnaires are commonly used because they can obtain standardised facts or views from many respondents in a relatively economical way. The same questions support comparison, tabulation and statistical analysis. They are suitable for respondents able to read and complete the form, but answers depend on respondents' understanding and cooperation. Good questionnaire research therefore requires clear items, proper response options, a covering letter, pilot testing, and attention to non-response.

3.4 Characteristics of a Good Questionnaire

The course gives **five characteristics**: it concerns a significant topic; requests information not readily available from other sources; is short enough to collect essential data; is attractive, neat and clear; and contains complete instructions, defined terms and simple items that express one idea each.

PAST QUESTION - 2024_2 Q2(a) – 8.75 marks

For a good questionnaire to serve the purpose which it is drafted for, it must contain five (5) basic characteristics, discuss these characteristics.

Answer: A good questionnaire (1) deals with a significant research topic clearly explained to respondents; (2) requests necessary information unavailable more easily elsewhere; (3) remains reasonably short and easy to complete; (4) has an attractive, orderly and legible layout; and (5) gives clear instructions, defines important terms and asks straightforward single-idea questions. Each characteristic improves response quality and helps reduce avoidable error.

3.5 Construction of a Questionnaire

The **eight steps** listed from Henderson and colleagues are: **(1)** identify objectives and required information; **(2)** select a response format; **(3)** determine respondents' frame of reference; **(4)** write items/questions; **(5)** prepare a data-summary sheet; **(6)** criticise, try out and revise questions; **(7)** assemble the questionnaire; **(8)** administer it. Keep the wording aligned with the intended users.

PAST QUESTION - 2025_1 Q1(b) – 12.5 marks

Describe the steps involved in the construction of a questionnaire.

Answer: Begin by identifying the research objectives and exactly what information is required. Choose an appropriate response format, then consider the language, experience and frame of reference of the target respondents. Write questions tied to each objective, plan how responses will be summarised, and critically review the items for clarity and relevance. Pilot-test and revise weak or confusing questions. Assemble the final questionnaire in a logical, attractive order with instructions before administering it to the chosen respondents. These are the eight development stages presented in the material.

3.6 Types of Questionnaires

Closed-ended/restricted questionnaires offer fixed answers (yes/no, categories or scales): easy to code, but limited in depth. **Open-ended/unrestricted questionnaires** allow respondents to explain in their own words: richer answers, but more demanding to complete, classify and analyse.

3.7 How to Improve Questionnaire Items

Use simple and direct wording; ask **one idea per item**; avoid slang, jargon and vague terms; define uncertain expressions; avoid double negatives and leading assumptions; supply adequate response alternatives; choose culturally appropriate wording; and ask only questions relevant to the study.

3.8–3.9 Administration and Pilot Testing

Before full administration, **pilot-test** on people resembling the target sample. The text gives **seven reasons**: measure completion time; check instructions and questions; delete unusable items; add missing items; estimate reliability; conduct item analysis; and confirm instrument usability. Then choose distribution and follow-up methods, and use a **covering letter** to explain the study's purpose and intended use of the information.

PAST QUESTION - 2024_2 Q2(b) – 8.75 marks

Highlight seven (7) reasons for conducting a pilot testing of the questionnaire.

Answer: (1) Estimate how long completion takes. (2) Confirm the clarity of items and instructions. (3) Remove questions that fail to yield usable data. (4) Add items needed to obtain relevant missing information. (5) Assess reliability. (6) Carry out item analysis. (7) Check overall usability before full administration. A pilot test reduces costly mistakes in the main survey.

Quick Revision: **Primary = original; secondary = existing. Questionnaire: five characteristics, eight construction steps, seven pilot-test reasons.**

UNIT 4: INTERVIEWS AND OBSERVATIONS

1.0 Introduction

Interviews and observation are useful when information cannot be obtained adequately from respondents filling in questionnaires alone.

3.1 What Is an Interview?

An **interview** is a direct oral interaction in which the researcher asks questions and records answers. It allows clarification and probing, although an interviewer must avoid influencing responses.

3.2 Types of Interview

Informal conversation: questions arise naturally during discussion. **Interview-guide approach:** topics are specified, while the interviewer chooses wording and order. **Standardised open-ended:** everyone receives predetermined questions in the same sequence, with freedom to answer. **Closed/fixed-response:** predetermined questions and alternatives are used, making comparison and analysis easy.

3.3–3.4 Arranging and Conducting an Interview

Select appropriate respondents, prepare and check questions, arrange a suitable time and place, establish rapport, explain the purpose, ask clearly and neutrally, probe relevant points and close courteously. The interviewer should listen carefully, avoid argument or leading questions, and preserve the accuracy of the response.

Strengths: personal contact, clarification, fuller explanations and participation by people who may not complete a written form. **Four recording methods:** classify responses into pre-set categories; summarise key points; take verbatim notes; or make audio/video recordings where acceptable.

3.5 Observation

Observation systematically watches and records the presence, absence or occurrence of verbal and non-verbal behaviour. The course distinguishes **participant** and **non-participant observation**. Relevant targets include **the setting, people, individual behaviour and group behaviour**.

3.6 Characteristics of a Good Observation

The source gives **seven standards**: careful planning; systematic attention to relevant events; awareness of the whole situation and significant details; objectivity and control of bias; separation of fact from interpretation; repetition or comparison for verification; and expert recording through suitable tools so the data are valid and reliable.

PAST QUESTION - 2024_1 Q2(a) – 8.75 marks

What are the seven (7) standard characteristics expected from an observer and their observations?

Answer: (1) Observations are carefully planned. (2) Observers work systematically and know what to look for. (3) They understand the whole situation while noticing significant details. (4) They strive for objectivity rather than personal bias. (5) They record facts separately from later interpretations. (6) Observations are checked through repetition or comparison with competent observers. (7) Observations are expertly recorded with suitable instruments to preserve valid, reliable data.

3.7–3.8 Uses and Recording of Observation

Observation helps examine actual behaviour, implementation of programmes and interactions that respondents may not describe accurately. Record **where and when**, who is present, what people say or do, the sequence of events and relevant context. Notes, schedules and recording devices should preserve facts before personal interpretation.

Quick Revision: **Four interview types; participant versus non-participant observation; seven observation standards.**

UNIT 5: PROBLEM IDENTIFICATION IN RESEARCH

1.0 Introduction

A research topic is only a broad area. The **research problem** is the specific unanswered issue within that area, stated so that a meaningful investigation is possible.

3.1 Concept of Problems in Research

A good problem may be expressed as a clear question about a relationship, difference, difficulty or unexplained occurrence. For example, asking whether poor learner support relates to low student participation is more researchable than simply writing 'learner support'.

3.2 Characteristics of Research Problems

A problem should be **clear, specific, researchable, supported by obtainable data, important and manageable**. It should imply questions or relationships that can be investigated rather than a purely personal opinion or an answer already known.

3.3 Some Sources of Suitable Research Problems

The material points to difficulties in schools, communities, banking, tourism and hospitality; changes in technology; professional experience; previous research; academic literature, abstracts and records; and gaps noticed while examining events. A good researcher turns an observed difficulty into a focused research question.

PAST QUESTION - 2022_2 Q2(A) – 15 marks

When conducting research, research is guided by certain sources to formulate a Research Topic. List and explain Five (5) Sources of Research Topics.

Answer: Five useful sources consistent with the course are (1) practical experience: recurring problems noticed in a workplace or community; (2) earlier studies and literature: gaps, recommendations and unanswered questions in published work; (3) institutional or social problems: difficulties in education, hospitality, banking and public services; (4) technological change: new systems that create questions about access, use or effectiveness; and (5) professional records and research abstracts: patterns and disagreements that suggest fresh inquiry. A source gives an idea; the researcher must still narrow it to a feasible, specific problem.

3.4 Selecting a Research Topic

Judge a topic by **personal and professional interest, relevance, availability of information, research skills, access to participants, time, funds and manageable scope**. Avoid choosing a topic simply because it sounds impressive. Check that it can be investigated scientifically and completed under the available conditions.

PAST QUESTION - 2023_1 Q2
Enumerate and comment on the guides in the choice of a research topic.

Answer: Consider (1) clear relevance to the field and an identifiable problem; (2) personal interest and the competence needed for the study; (3) availability and accessibility of dependable data; (4) novelty or a gap in existing research; (5) realistic time, money and logistical demands; (6) ethical acceptability and permission to access participants; and (7) an achievable scope, defined population and suitable method. These guides help select an important study that can actually be completed.

3.5 Shaping and Refining the Problem

Move from a broad topic to a precise issue by reviewing related work, delimiting the population and location, deciding the major variables, writing clear objectives and research questions, and checking feasibility. A refined problem tells the reader **what is being investigated, among whom and why**.

Quick Revision: **Topic is broad; problem is specific.** A good problem is significant, researchable and feasible.

MODULE TWO

UNIT 1: STEPS IN RESEARCH AGENDA

1.0 Introduction

After identifying a research problem, the investigator sets the study's boundaries, explains key terms, formulates hypotheses, identifies variables and checks whether the investigation is feasible.

3.1 Components of Setting the Problem

Delimitations are boundaries deliberately fixed by the researcher, such as area, population, period or variables studied. **Limitations** are restrictions not fully under the researcher's control, such as unavailable records, participants who do not respond or an instrument's weakness. **Assumptions** are statements taken as true for conducting the study even though they cannot all be verified. **Significance** states who benefits from the findings and how.

PAST QUESTION - 2025_2 Q3 – 15 marks; 2024_2 Q3(b) – 8.75 marks
Differentiate between limitations and delimitations in research.

Answer: Delimitations are intentional boundaries imposed by the researcher to make the study specific and manageable: choosing a particular study centre, limiting the period to one semester or investigating only 300-level students. Limitations are difficulties or conditions beyond the researcher's full control that restrict procedures or interpretation: poor access to records, non-response, inadequate instruments or inability to randomly assign participants. Delimitations explain what the researcher chooses to include or exclude; limitations explain what may weaken or restrict the conclusions despite those choices. Both should be disclosed so readers understand the scope and strength of the study.

3.1.4 Definition of Terms

Operational definitions give an unusual or important term the precise meaning it has **within this particular study**. For example, 'regular attendance' might be defined in the research as attending a specified number of sessions, allowing another researcher to identify how it was assessed. The definition reduces ambiguity, guides measurements and makes the findings easier to evaluate or replicate.

PAST QUESTION - 2025_1 Q2(a) – 7.5 marks; 2023_1 Q1(a)
Explain the significance of operational definition of terms in research.

Answer: Operational definition states exactly how an important concept is understood or recognised in a given investigation. It (1) prevents several interpretations of one term; (2) fixes the researcher's frame of reference; (3) shows how variables and outcomes are identified or measured; (4) helps readers judge whether the objectives were achieved; and (5) improves replication and comparison. An operational definition is study-specific, not merely a general dictionary meaning.

3.2 Hypothesis

A **hypothesis** is a tentative, intelligent guess about how a problem may be explained or how variables may relate. It directs data gathering, helps determine measurements, focuses analysis and provides a proposition to test. **Null hypothesis (H0):** ordinarily states that no significant difference or relationship exists, e.g. no significant difference in spending patterns. **Alternative hypothesis (H1):** proposes a difference or relationship, possibly indicating its direction.

A **researchable hypothesis** should be **clear, specific, relevant to the stated problem, logically consistent, expressed in terms of identifiable variables and capable of empirical testing**. State it so available evidence can support or fail to support it.

PAST QUESTION - 2022_2 Q1(A) – 10 marks
Hypotheses are regarded as tentative answers to research problems. They are usually expressed in the form of a relationship between independent and dependent variables. Expatiate on research hypotheses and its characteristics with examples.

Answer: A research hypothesis is a provisional, testable explanation or prediction offered as a possible answer to a research problem. It usually states an expected relation between variables, thereby guiding the data required and its analysis. A good hypothesis is clear, specific, logically grounded, relevant, measurable or otherwise testable, and open to confirmation or rejection by evidence. Example: 'Increased funding improves the quality of hotel services' connects funding (independent variable) with service quality (dependent variable). Its null form is: 'There is no significant relationship between funding and service quality.' A hypothesis is not a final conclusion.

PAST QUESTION - 2023_1 Q1(b)
State the characteristic of a Researchable Hypothesis.

Answer: A researchable hypothesis must be clear and precise, linked to a defined research problem, concerned with identifiable variables, logically justifiable, possible to investigate with available data, and stated so evidence can support or refute it. It should not be vague, self-evident or impossible to test.

PAST QUESTION - 2025_1 Q2(b) – 7.5 marks
Discuss the role of hypotheses in the research process.

Answer: Hypotheses turn the research problem into specific propositions for investigation. They define relationships to be examined, direct what data and instruments are needed, help determine the research design, guide the choice of analysis, and provide a basis for interpreting findings. Their test results assist the researcher in answering the original question. A hypothesis guides inquiry; it does not predetermine the answer.

3.3 Variables

A **variable** is a quality that can take different values. An **independent variable** is treated as the presumed cause or influence; the **dependent variable** is the presumed effect or outcome. **Discrete variables** have distinct countable values (e.g. family size); **continuous variables** can take measurements across a range (e.g. age or scores). The course also mentions extraneous, intervening and moderating variables.

3.4 Feasibility of a Research Problem

Before beginning, evaluate **time** needed for instrument construction, sampling, data collection and analysis; **finance** for printing, travel, assistants and processing; and **practical risks** such as access and administrative difficulties. A pilot study helps identify weaknesses. If needed, reduce the sample, scope or number of hypotheses to what can be completed.

Steps in the Research Process

The material gives an extended sequence: **identify the problem; develop hypotheses; identify and label variables; define key terms operationally; manipulate/control variables when appropriate; design the study; construct instruments and interview schedules; select the sample; collect data; analyse it; discuss findings; draw conclusions and recommendations; and write the report**. Some exam papers request a shorter number of stages; the following answer groups the source's steps without changing their logical order.

PAST QUESTION - 2023_1 Q1(c); 2024_2 Q3(a) – 8.75 marks; 2022_2 Q5(A) – 15 marks
Discuss the activities or processes involved in conducting research; explain seven processes involved in research.

Answer: The research process may be summarised in seven grouped stages from the course's longer sequence: (1) identify, state and refine the research problem; (2) review existing information and formulate questions or hypotheses; (3) identify variables, operationally define terms and choose an appropriate design; (4) select the population/sample and prepare or validate instruments; (5) gather the relevant data; (6) organise and analyse the data, test the hypotheses and interpret the findings; and (7) state conclusions and recommendations and write the research report. The source itself gives more detailed individual activities, so seven is a useful grouping rather than an additional official list.

Quick Revision: **Delimitation = chosen boundary; limitation = unavoidable constraint. H0 = no significant difference; H1 = proposed difference.**

UNIT 2: THE TYPOLOGIES OF THE RESEARCH METHODS I

1.0 Introduction

This unit compares empirical, phenomenological and critical approaches, showing that research designs are chosen according to the sort of problem being investigated.

3.1 Empiricism

Empiricism treats sensory experience, observation and measurement as a central basis of knowledge. Empirical inquiry begins with a question and uses evidence to build or test explanations. The source's empirical process includes **data collection; measurement; classification, organisation and tabulation; analysis and reduction; formulation of hypotheses; and formulation of theory or law**. Its strength lies in systematic, open-to-check evidence.

3.2 Phenomenology

Phenomenology seeks to understand a phenomenon as experienced and given meaning by participants. Rather than start with numerical measurement alone, it explores personal perceptions, intentions and everyday social experience. The material discusses approaches such as **ethnomethodology** and interpretation of participants' accounts; the researcher attends to the meanings people attach to their actions.

3.3 Critical Research

Critical social research questions social arrangements, dominant ideas, contradictions and unequal interests instead of merely recording observable conditions. It examines how institutions and social structures shape experience and how apparently accepted explanations may conceal conflict.

3.3.3 Approaches in Critical Social Research

Critical case study conducts detailed investigation of a case to uncover a specific contradiction or myth, often combining structured interviews and observation. **Radical historicism** interprets historical evidence in relation to prevailing social structures, looking beneath the surface account of events. **Critical ethnography** uses close observation and social/historical analysis to expose taken-for-granted practices and contradictions. **Structuralist techniques** examine signs, meanings, binary oppositions and narrative sequences.

PAST QUESTION - 2024_1 Q2(b) – 8.75 marks

Write short notes on the following approaches in critical social research: (i) critical case study (ii) radical historicism (iii) critical ethnography.

Answer: (i) Critical case study: detailed study of a selected case to uncover hidden interests, myths or contradictions, using interviews, observation and other evidence. (ii) Radical historicism: interprets the historical development of structures by connecting evidence of past events with dominant social arrangements rather than merely listing dates and facts. (iii) Critical ethnography: closely studies ordinary group behaviour and culture to reveal unquestioned assumptions, contradictions and the wider social or historical forces behind them. Each approach aims to explain, not simply describe, accepted social practices.

Quick Revision: **Empirical = observable evidence; phenomenological = lived meaning; critical = examination of social contradictions and structures.**

UNIT 3: PHILOSOPHICAL AND HISTORICAL RESEARCH

1.0 Introduction

Both approaches investigate questions that cannot always be settled by immediate controlled experiment. Philosophical inquiry examines meanings and arguments; historical inquiry investigates past events through evidence.

3.1 Philosophical Research

Philosophical research critically and logically examines concepts, values, assumptions and systems of thought. Its focus is sound reasoning and justified interpretation. It may investigate issues such as equality of opportunity or the underlying ideas of an educational policy.

PAST QUESTION - 2024_1 Q3(a) – 8.75 marks
Explain the concept of philosophical research.

Answer: Philosophical research is systematic, critical examination of ideas, concepts, assumptions and values. It uses logical analysis, comparison, argument and interpretation to clarify the meaning and justification of a position or system of thought. Unlike a survey that chiefly counts responses, it studies the coherence and implications of ideas, using relevant philosophical texts and other documentary material.

3.1.1 Main Steps in Philosophical Inquiry

The material sets out **topic identification** (choose a meaningful, justifiable question and consult existing writing); **data collection** (obtain relevant authentic books, philosophical works, records and commentaries); **classification and interpretation** (organise ideas, compare, appraise, cross-examine and synthesise them); and **drawing and presenting justified conclusions**. Personal bias should be controlled during interpretation.

PAST QUESTION - 2024_1 Q3(b) – 8.75 marks
Outline the main steps in philosophical enquiry.

Answer: (1) Select an appropriate philosophical topic capable of reasoned examination. (2) Study existing literature and identify the questions to be addressed. (3) Collect relevant and authentic writings, commentaries and records. (4) Classify material under meaningful themes and examine competing views. (5) Interpret through description, comparison, critical appraisal and logical synthesis. (6) Present conclusions supported by the analysed arguments, limiting subjective bias.

3.2 Historical Research

Historical research systematically locates, evaluates and synthesises evidence of earlier events to establish facts, explain developments and understand their significance for present conditions. The material recognises **primary historical sources**, created close to the event, and **secondary sources**, which interpret or describe earlier records.

3.2.1–3.2.2 Features and Main Steps in Historical Research

Historical research relies on **past evidence, careful source criticism, chronological/contextual understanding, reconstruction of events and cautious interpretation**. Steps: **identify a historically significant problem; specify relevant data; collect records and discover new sources; organise material; critically evaluate and interpret the evidence; and report the account and implications**. Researchers check where, when, by whom and under what circumstances evidence was produced.

Quick Revision: **Philosophical inquiry tests ideas and arguments; historical inquiry tests evidence about the past.**

UNIT 4: NATURALISTIC INQUIRY AND CASE STUDY

1.0 Introduction

Naturalistic inquiry and case study examine events in real settings and aim at deep understanding rather than artificial control of every condition.

3.1 Naturalistic Inquiry

Naturalistic inquiry studies behaviour and meaning in the environment where they ordinarily occur. Its five distinctive qualities in the text are **holistic approach, insightful inquiry, absence of an a priori theory, no pre-specified rigid design, and naturalistic setting**. Investigators may use interviews, observation, diaries, records and sometimes quantitative information.

3.2 Main Steps in Naturalistic Inquiry

Start with broad field questions, enter and learn about the setting, gather detailed accounts, record field observations, check emerging interpretations and revisit the data as questions evolve. The course describes the process as cyclical, involving observation, tentative explanations, further examination and refinement.

3.2.1 Trustworthiness and Objectivity

Credibility asks whether interpretations reflect respondents' realities. **Transferability** asks whether results may offer useful understanding in another suitable context. **Dependability** examines stability of the information and interpretations. **Confirmability** asks whether independent scrutiny of evidence could reach comparable conclusions.

3.3 Case Study Research

A **case study** is a deep, sustained examination of a particular person, institution, group or situation, often using multiple sources of evidence. The course lists **five characteristics: continuity in investigation, completeness, authenticity of data, confidential recording and intellectual synthesis**. It is especially suitable for complex practical situations that require a complete picture.

PAST QUESTION - 2024_2 Q4(a) – 8.75 marks
Briefly explicate five (5) characteristics of a case study method.

Answer: (1) Continuity: inquiry is sustained long enough to uncover underlying factors. (2) Completeness: information is gathered from the case's relevant internal and external setting. (3) Authenticity: evidence must be dependable and appropriately checked. (4) Confidential recording: sensitive case data are systematically recorded and protected. (5) Intellectual synthesis: diverse observations, interviews and records are brought together into a coherent explanation of the case.

3.4 Main Steps in Case Study

Select and define the case and obtain preliminary background; **collect first-round data** through interviews, observation, records or tests; **analyse the initial evidence** to locate unresolved factors; **conduct a focused second-round investigation**; **interpret and synthesise** the information; and where the case requires intervention, **introduce and assess alternative measures**. Cases can generate hypotheses and a context-specific explanation, even if generalisation is limited.

PAST QUESTION - 2025_2 Q4 – 15 marks; 2024_2 Q4(b) – 8.75 marks

Describe the process of conducting a case study research; discuss the concept of case study research and state the main steps in a case study.

Answer: Case study is intensive investigation of one bounded case in its actual context. First, select a person, institution or situation and clarify why it requires investigation. Establish background information and collect initial data by interview, observation, documents or other appropriate instruments. Analyse these findings to identify influential factors and questions requiring deeper inquiry. Carry out a second, more focused round of data gathering, cross-check accounts and interpret the evidence as a whole. Where the purpose is remedial, propose or test appropriate alternative measures. The final account should preserve confidentiality, explain the case coherently and avoid claiming wider generalisation unsupported by the evidence.

3.4.1 Is Case Study Scientific in Nature?

The text recognises criticisms that a single case may not support broad generalisation. Nevertheless, carefully recorded and checked case evidence can support meaningful hypotheses, qualitative testing and insights that guide later studies. Scientific value depends on systematic data and disciplined interpretation.

Quick Revision: **Naturalistic trustworthiness = credibility, transferability, dependability, confirmability. Case study = five characteristics and staged investigation.**

UNIT 5: DESCRIPTIVE RESEARCH

1.0 Introduction

Descriptive research studies and interprets existing conditions: what is happening, where it occurs and how characteristics are distributed. It does not necessarily require manipulating variables.

3.1 Introduction to Descriptive Research

The approach systematically **describes, records, analyses and interprets existing conditions, practices and relationships**. It is useful in education, organisations, government, tourism and industry when reliable information about current status is required.

3.2 Main Steps in Descriptive Research

Define the problem and purpose; identify the relevant population, variables and information needed; select a descriptive design and sample; prepare suitable observation, interview or questionnaire instruments; collect and organise data; analyse the findings; and interpret, conclude and report. Control bias in sampling, instrument wording and interpretation.

3.3 Types of Descriptive Research

Survey studies: obtain existing-status information from a whole population or representative sample; data may show current status, comparisons with a standard or ways to improve conditions. **Analysis of documents:** examines existing records, texts and reports for relevant evidence. **Correlational studies:** estimate relationships between variables without establishing causality. **Causal-comparative (ex post facto) studies:** compare already-existing groups or conditions to seek plausible explanatory factors without experimental manipulation.

3.3.1 Survey Study: Practical Stages

A survey involves **planning** (problem, population, methods and procedures); **sampling** when full coverage is not realistic; **preparing and pilot-testing a data instrument**; **collecting information consistently**; **tabulating and analysing responses**; and **reporting the findings with appropriate limits**. Surveys require representative samples and straightforward questions for valid comparison.

PAST QUESTION - 2025_1 Q3 – 15 marks

Explain the main steps involved in conducting a survey research.

Answer: (1) Identify a clear issue that requires information on a population's existing situation. (2) State objectives, questions and the data needed. (3) Define the target population and select an appropriate full-coverage or sampling plan. (4) Choose and construct an instrument such as a questionnaire or interview schedule. (5) Pilot-test items and correct weaknesses. (6) Administer the instrument systematically and address non-response. (7) Edit, classify and tabulate the returns. (8) Analyse the data to answer the questions. (9) Interpret results, state limitations and present conclusions and recommendations. These stages follow the general descriptive/survey procedure explained in the material.

Quick Revision: **Descriptive = describes what exists. Survey, document analysis, correlation and causal-comparative are its main categories.**

MODULE THREE

UNIT 1: EXPERIMENTAL AND ACTION RESEARCHES

1.0 Introduction

Experimental research studies the effect of a deliberately introduced change. Action research focuses on a practical local problem and evaluates an intervention intended to improve the situation.

3.1 Experimental Research

Experimental research is inquiry in which the researcher manipulates an **independent variable**, controls relevant surrounding factors and measures the resulting changes in a **dependent variable**. The central question is whether the treatment produced the observed effect. Comparison groups and appropriate assignment help reduce alternative explanations.

3.2 Characteristics of Experimental Research

The course stresses **control**, **manipulation** and **observation**. **Control** reduces or isolates extraneous influences; **manipulation** deliberately changes a treatment or independent variable; **observation** measures the dependent variable to determine the effect. Careful planning and, in true experiments, random assignment support stronger causal conclusions.

PAST QUESTION - 2025_1 Q4 – 15 marks; 2024_1 Q4(a) – 8.75 marks
Define Experimental Research and discuss its main characteristics; briefly explain the concept and three characteristics of experimental research.

Answer: Experimental research is systematic investigation of a possible cause-and-effect relationship by deliberately varying an independent variable and observing an outcome while other relevant conditions are controlled. Its three principal characteristics are control (hold constant or minimise influences outside the treatment), manipulation (introduce different treatment conditions) and observation (measure responses or dependent-variable changes after the treatment). For example, a researcher may compare customer-service performance after two different staff-training arrangements, while controlling relevant conditions. Strong experiments also describe their participants, instruments and method of comparison clearly.

3.3 Steps in Experimental Research

The source identifies **survey of related literature; problem identification and definition; hypothesis formulation and definition of terms and variables; and construction of an experimental plan**. The plan specifies extraneous variables and their control, design, sampling and allocation to groups, valid measuring instruments, data-collection procedures and any pilot study. Carry out the treatment, gather results, analyse statistically and test whether differences are significant.

PAST QUESTION - 2025_2 Q5 – 15 marks
Describe the steps involved in conducting Experimental Research.

Answer: (1) Examine relevant previous literature and identify a problem suitable for an experiment. (2) State the problem, hypotheses, variables and operational definitions. (3) Choose an appropriate experimental design. (4) Identify possible extraneous variables and plan controls. (5) Select subjects and, where the design requires it, assign them to treatment and comparison groups. (6) Construct or select valid outcome measures and pilot-test procedures when necessary. (7) Administer the planned treatment and collect observations in a consistent way. (8) Organise and analyse the results statistically, test the hypothesis and state supported conclusions. This order follows the preparation, conduct and analysis stages of the supplied material.

3.4 Experimental Designs

Pre-experimental designs offer limited control or comparison. **True experiments** normally use random assignment and stronger controls. **Quasi-experiments** investigate effects when random assignment or full control is unavailable. **Factorial designs** examine more than one independent variable or treatment factor. **Time-series designs** make repeated measurements to study changes across time. A design must suit the problem and practical conditions.

3.5 Action Research

Action research addresses an immediate problem within a particular institution, group or setting. Practitioners and interested parties may participate in identifying the difficulty and assessing a possible remedy. Its emphasis is improvement through planned action and reflection, rather than broad statistical generalisation.

3.6 Stages of Action Research

The source gives **four stages**: **(1) Diagnosis** (identify, evaluate, discuss and refine the underlying problem); **(2) Planning and intervention design** (select procedures, sample, treatment and evaluation methods); **(3) Intervention and impact assessment** (carry out the action, gather and interpret evidence of change); and **(4) Reflection** (consider why the intervention did or did not work and determine the next steps).

PAST QUESTION - 2024_1 Q4(b) – 8.75 marks

Mention four stages involved in the conduct of action research.

Answer: The four stages are diagnosis, which clarifies the real underlying problem; planning and intervention, which designs an appropriate response and evaluation procedure; intervention and impact assessment, which implements the action and assesses the resulting evidence; and reflection, which interprets the results and determines whether further improvement or a revised action is needed.

Quick Revision: **Experimental = control + manipulation + observation. Action research = diagnose, plan, intervene/evaluate, reflect.**

UNIT 2: POPULATION AND SAMPLING

1.0 Introduction

Researchers usually cannot investigate everybody in a large population. Sampling provides a practical way to obtain information from a smaller number of suitable members.

3.1 Population and Samples

A **population** is the entire set of persons or items sharing characteristics relevant to a study. A **sample** is the part chosen from that population for investigation. A population is not simply the geographical study area. An appropriate sample should represent the relevant characteristics of the target group as far as the selected design permits.

3.2 The Purpose of Sampling

Sampling reduces costs, time and fieldwork, makes detailed investigation manageable and permits conclusions about a wider population when selection procedures justify such inference. Poor selection may introduce sampling bias.

3.3 Types and Procedures for Sampling

Probability sampling: each population element has a known, non-zero chance under a random selection procedure; it supports stronger statistical generalisation and estimation of sampling error. **Non-probability sampling:** cases are selected through convenience, judgment, fixed categories or referrals rather than known random probabilities; it can be practical but generalisability is less secure.

PAST QUESTION - 2025_1 Q5 – 15 marks

Differentiate between Probability and Non-Probability Sampling techniques.

Answer: In probability sampling, selection follows a chance procedure and each element has a known probability of inclusion. Techniques include simple random, systematic, stratified random and cluster sampling. It usually improves representativeness, supports statistical inference and permits estimation of sampling error, but may require a suitable population list and more resources. In non-probability sampling, the researcher selects respondents without known inclusion chances, using accessibility, judgment, quotas or referrals. It can be cheaper and suitable for exploratory or hard-to-reach groups, but selection bias is harder to assess and findings should not automatically be generalised statistically to the whole population. The essential distinction is random selection with known chances versus non-random selection without known chances.

3.4 Sampling Techniques

Simple random sampling: every eligible unit is selected through a random procedure. **Systematic sampling:** after a random start, take every kth element from an ordered list. **Stratified random sampling:** divide a heterogeneous population into relevant strata and sample randomly within each. **Cluster sampling:** choose naturally occurring groups, then study units in selected clusters. **Non-probability methods:** choose participants using convenience, purposive/judgment, quota or comparable non-random procedures.

3.5 Sample Size

Sample size depends on the population, the accuracy required, variability of characteristics, design, time and available resources. A larger sample is not automatically a better one if it is selected with bias. Describe clearly who is eligible and how the final sample is obtained.

Quick Revision: **Population = whole target group; sample = part selected. Probability = known chance; non-probability = no known random chance.**

UNIT 3: STATISTICAL TOOLS OF ANALYSIS

1.0 Introduction

Once data have been collected, the researcher organises them, chooses suitable statistics, interprets the results and links them back to the research questions.

3.1 Data Analysis: Definition and Meaning

Data analysis means classifying, organising, summarising and examining the collected information to answer research questions or test hypotheses. Tables, frequencies, percentages and charts help present data; statistical techniques help interpret relationships, differences and variation.

PAST QUESTION - 2022_2 Q1(B) – 8 marks; 2023_1 Q3(a)
Discuss the importance of data analysis in research; explain the term data analysis.

Answer: Data analysis is the systematic organisation, summarisation and interpretation of research data so that meaningful findings can be drawn. It transforms raw responses into understandable tables, distributions or measures; reveals patterns and differences; supports hypothesis testing; checks the implications of collected evidence; and provides a basis for findings, conclusions and recommendations. Without appropriate analysis, a large collection of figures or interview responses may not answer the original research question.

3.2 What Is Statistics?

Statistics provides procedures for collecting, presenting, analysing and interpreting numerical information. **Descriptive statistics** summarise observed data; **inferential statistics** help assess hypotheses or make carefully qualified conclusions beyond the immediate sample.

3.3 Parametric and Non-Parametric Statistics

Parametric procedures generally depend on assumptions about the population distribution and measurement scale. **Non-parametric procedures** are used when such assumptions are unsuitable or when the form of data calls for different tests. Select methods for the research question and actual characteristics of the data.

3.4 Measures of Central Tendency

Mean = sum of all scores divided by the number of scores. **Median** = middle value after ordering the scores (or the average of the two middle values if the count is even). **Mode** = most frequently occurring value. The mean uses all values but is sensitive to extremes; the median is positional; the mode identifies the commonest score.

PAST QUESTION - 2023_1 Q5

A group of 9 students had Mathematics scores: 70, 65, 60, 55, 75, 80, 85, 90, 95. Find the mean and median.

Answer: Total score = 70 + 65 + 60 + 55 + 75 + 80 + 85 + 90 + 95 = 675. Mean = 675 ÷ 9 = 75. Ordered scores = 55, 60, 65, 70, 75, 80, 85, 90, 95. The fifth observation is the middle of nine scores, so median = 75.

PAST QUESTION - 2024_2 Q5(a) – 8.75 marks

Calculate the mode of the following sets of numbers: 1, 2, 3, 4, 5, 6, 6, 7, 8, 9.

Answer: Mode = 6, because six occurs twice while every other number occurs once. The mode is the value with the highest frequency.

3.5 Measures of Variability

Range is the difference between the highest and lowest scores. **Mean deviation** summarises average absolute departure from a central value. **Variance** summarises squared deviations; **standard deviation** is the square root of variance and expresses spread in the original measurement units. These show how tightly or widely values are distributed.

3.6 The Normal Curve

The **normal curve** is bell-shaped, highest at the centre and symmetrical about its vertical axis. **Mean = median = mode** in the ideal curve. Scores cluster towards the centre; the curve extends in both directions, approaching but not cutting the baseline. It represents probabilities, not certainty. Uses noted in the material include percentile ranks, normalising distributions and testing significance.

PAST QUESTION - 2024_2 Q5(b) – 8.75 marks

Outline six (6) features of a normal curve.

Answer: (1) It is symmetrical around the central vertical axis. (2) Values cluster near the centre. (3) Mean, median and mode coincide. (4) The curve is highest near its mean. (5) Its tails extend indefinitely in both directions and approach the baseline. (6) It represents a probability distribution, not a guarantee that any single outcome must occur. These reflect the course's list of normal-curve features.

3.7 Measures of Association

Measures of association examine **how two variables vary together**. A positive relationship means their values tend to increase together; a negative one means they tend to move in opposite directions. A relationship does not automatically establish cause and effect.

3.8–3.9 Parametric and Non-Parametric Tests

The course discusses testing **H0** with a chosen **level of significance** and making a decision from sample evidence. A significance level represents the error risk set for the statistical decision. Parametric and non-parametric tests are selected according to the data and their assumptions. Report what the evidence supports, not what the researcher hoped to prove.

PAST QUESTION - 2022_2 Q1(C) – 2 marks
List two software that could assist researchers to analyse collected data.

Answer: The supplied FMS 304 course material does not name any statistical software packages. Therefore, a two-name answer cannot be established from the required source alone. The question is retained here for exam awareness without adding unsupported material.

Quick Revision: **Mean = total ÷ count; median = ordered middle; mode = highest frequency.** Normal curve: symmetric; mean = median = mode.

UNIT 4: WRITING RESEARCH REPORTS AND PROPOSALS

1.0 Introduction

A good investigation should be communicated in a clear report that lets readers understand the problem, the procedures used, the evidence found and the conclusions reached.

3.1 Chapters and Sections Constituting a Research Project Report

The material organises the report into **preliminary pages**, **Chapter One: Introduction**, **Chapter Two: Literature Review**, **Chapter Three: Methodology**, **Chapter Four: Presentation and Analysis of Results**, **Chapter Five: Discussion, Summary, Conclusions and Recommendations**, followed by supplementary pages and references as applicable.

3.2 The Preliminary Pages

The **title page** identifies the topic and research context. Other pages may include **certification or approval, dedication, acknowledgements, abstract and table of contents**, along with relevant lists. An abstract briefly indicates the problem, method, major findings and conclusions.

3.3 Chapter One – The Introduction

This sets out the **background**, **statement of problem**, **purpose/objectives**, **research questions or hypotheses**, **significance**, **scope/delimitations**, **limitations** and **definitions of terms** as appropriate. It should make clear why the problem deserves investigation and exactly what the study intends to accomplish.

PAST QUESTION - 2022_2 Q4(c) – 5 marks
Explain the term Research Objectives.

Answer: Research objectives are clear statements of what the investigation intends to achieve. They translate the general research problem into specific tasks or results, guide data collection and analysis, and provide criteria for evaluating the study's success. Objectives should be relevant, precise, achievable and consistent with the questions or hypotheses.

3.4–3.5 Chapter Two – Literature Review

A **literature review** critically examines existing conceptual, theoretical and empirical writing related to the study. It identifies what scholars have found, where they disagree, what has not yet been adequately examined and how the proposed study fits existing knowledge. It is not simply a list of books or copied quotations. Organise by useful themes and cite sources properly.

PAST QUESTION - 2022_2 Q4(a) – 5 marks; 2023_1 Q6

Explain the term Literature Review; comment on the statement that a literature review in scientific research examines recent research studies.

Answer: A literature review is an organised, critical examination of earlier and relevant studies, theories and concepts relating to a research problem. It identifies established findings, methods used, inconsistencies and gaps, thereby justifying the new investigation and informing its design. Recent studies are important because they show current evidence and debates, but the review may also require influential older works. The aim is relevant critical synthesis, not merely a chronology of recent publications.

3.5 Chapter Three – Methodology

Research methodology explains the plan and actual procedures: the **research design**, target **population**, **sample and selection procedure**, **instrument(s)**, validation and reliability where appropriate, **data-collection steps**, controls and the **analysis method**. State the process sufficiently clearly that another researcher can understand or evaluate it.

PAST QUESTION - 2022_2 Q4(b) – 5 marks

Explain the term The Research Design and Methodology.

Answer: Research design is the organised plan showing how the research problem will be investigated, how variables are handled and how evidence will be obtained and assessed. Methodology explains the practical procedures: population and sample, sampling technique, instrument preparation, validity and reliability, data collection and data analysis. Design states the structure of the inquiry; methodology reports how that structure is put into practice.

3.6 Data Analysis and Presentation of Results

Results are set out clearly with suitable tables, figures or narrative summaries. Each presentation should answer a research question or address a hypothesis; the researcher should interpret correctly and distinguish observations from speculation.

3.7 The Discussion

Discuss the meaning of findings in relation to objectives, hypotheses, reviewed literature and possible explanations. Where results differ from earlier findings, explain the disagreement only as far as the evidence justifies.

3.8 Summary, Conclusion, Recommendations and Implications

Summary briefly restates purpose, method and major results. **Conclusion** draws the central interpretation supported by data. **Recommendations** suggest practical action or further study. **Implications** explain what the findings mean for practice, theory or policy. **Supplementary pages** may contain references, instruments, supporting tables and appendices.

Quick Revision: **Chapter 1 = problem; 2 = literature; 3 = method; 4 = results; 5 = meaning and conclusions.**

UNIT 5: RESEARCH REFERENCING, PROPOSAL CONTENTS AND ETHNICAL ISSUES

1.0 Introduction

A final research work should acknowledge its sources, make research intentions clear when still at proposal stage and protect the people and institutions whose information is used.

3.1 Research Referencing

Referencing lists the works actually cited so readers can locate and verify them and so the author gives proper credit. Check the accuracy of names, date, title, publisher, place and page numbers where needed. The course's categories are **books; articles/chapters in books; government documents; periodicals; and other material**.

3.1.1–3.1.5 References from Different Sources

Books: surname, initials, year, title, place and publisher (according to the pattern taught in the material). **Book chapters:** chapter author/year/title, 'In' editor, book title, publisher and chapter pages. **Government documents:** issuing institution, year, title and publication details. **Periodicals:** author, year, article title, periodical title, volume/issue and pages. **Others:** technical reports, conference papers, dissertations, manuscripts, personal communications and minutes, with the relevant identifying details. **Personal communications** are cited in the text by communicator and date in the material's illustration.

3.2 Proposal Contents

A **research proposal** is prepared **before** the investigation and explains what will be studied, why and how. The course identifies six possible components: **(1) Chapter One – Introduction, (2) Chapter Two – Literature Review, (3) Chapter Three – Methodology, (4) Time Schedule, (5) Proposal Budget and (6) Supplementary Pages, where applicable**. The main three chapters constitute the normal degree proposal; **time schedules and budgets apply especially to grant or contract proposals**, not ordinarily to school supervisor/defence proposals. Use **future/intended-action language** in a proposal; a completed report describes actions already undertaken, commonly in past tense.

PAST QUESTION - 2024_1 Q5(a) – 8.75 marks; 2022_2 Q3(A) – 15 marks
Explicate the six main components of a research proposal; discuss the components of a comprehensive Research Proposal.

Answer: The six components in the NOUN material are (1) Chapter One – Introduction: background, problem, objectives, questions/hypotheses, significance and scope; (2) Chapter Two – Literature Review: relevant theories, concepts, studies and gaps; (3) Chapter Three – Methodology: proposed design, population, sample, instruments, data collection and analysis; (4) Time Schedule: expected timing of major research activities; (5) Proposal Budget: estimated expenses and resources; and (6) Supplementary Pages: relevant references and appendices. For an ordinary degree proposal, the first three chapters form the core; the course specifically says time schedules and budgets are generally associated with grants/contracts, not the normal supervisor or faculty-defence proposal.

3.3 Ethnical (Ethical) Issues in Research

The course uses the heading '**Ethnical Issues**' while explaining **research ethics**. Its key expectations include **informing participants** of objectives, procedures and possible uses; obtaining **free informed consent**; respecting voluntary participation; protecting **confidentiality** and sensitive institutional records; seeking **institutional permission** before using private data; avoiding unjustified **physical, emotional, social, political or financial harm**; and acting with **truthfulness, honesty and objectivity** in collection and reporting.

PAST QUESTION - 2024_1 Q5(b) – 8.75 marks

Describe six ethical issues in research.

Answer: (1) Informed consent: explain the purpose and procedures and obtain voluntary agreement. (2) Freedom to participate: do not coerce respondents or violate free informed choice. (3) Confidentiality: protect sensitive personal and institutional information. (4) Permission for institutional data: seek prior approval before accessing or publishing information supplied for restricted purposes. (5) Protection from harm: avoid unjustified physical, emotional, psychological, political, moral or economic risks. (6) Honesty and objectivity: collect and present information truthfully and distinguish internal monitoring findings from results authorised for public release.

Proposal Versus Completed Report

Proposal: tells what the researcher **will** investigate; usually presents the first three chapters and planned methods. **Final report:** tells what the researcher **did**, the results obtained, what they mean and the conclusions drawn. Both must use appropriate referencing and follow ethical requirements.

Quick Revision: **Proposal = planned work/future tense; report = completed work/results. Ethical research protects consent, confidentiality, permission, safety and truthfulness.**

EXAMINATION STUDY CHECK

Across the supplied papers, pay particular attention to **research characteristics and purposes; the four scientific assumptions; hypothesis versus theory; questionnaire construction, qualities and pilot testing; definitions of terms; delimitations versus limitations; observational standards; qualitative and case-study procedures; descriptive/survey research; experimental and action research; sampling; mean/median/mode and normal curve; proposal components; and ethical issues**.

The two copies of the **2024_1** paper were identical and have been counted as one session. The 2023_1 paper carries a different course-title wording ('Research Methods for Business Decisions'), but the course code is FMS 304 and the supplied 2023 NOUN course material is titled 'Research Methodology'. For consistency this summary follows the material's title and structure.

SOURCE NOTE: Prepared strictly from the supplied FMS 304 NOUN course material and six distinct examination sessions (seven files). Where a paper requests information absent from the material, that gap is identified rather than filled from outside sources.

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